



GE Medical Systems

gemedical.com

Technical Publication

Direction 5117718-100

Revision 2

GE Medical Systems

**LightSpeed 5.X Plus Software Installation
Procedures**

Copyright © 2004-2005 by General Electric Company, Inc.
All Rights Reserved.



LEGAL NOTES

TRADEMARKS

Adobe, the Adobe logo, Acrobat, the Acrobat logo, Exchange, and PostScript are trademarks of Adobe Systems Incorporated or its subsidiaries and may be registered in certain jurisdictions.

Microsoft is a registered trademark and Windows is a trademark of Microsoft Corporation.

All other products and their name brands are trademarks of their respective holders.

COPYRIGHTS

All Material Copyright © 2004 by General Electric Company, Inc. All rights reserved.

IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓ LO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

このサービスマニュアルには英語版しかありません。

GE MS 以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。

このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。

この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意:

本维修手册仅存有英文本。

非 GE MS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage that may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION
Risk of
Explosion

ATTENTION
Danger
d'Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

End of Section

Revision History

Revision	Date	Reason for change
2	01/26/05	Deleted former Chapter1 and Appendix B. Updated DVD Interchange Software Installation (current Chapter 1). Added patch software installation procedures. Added Class C and Class M installation procedures.
1	08/26/04	Initial release document.

Table of Contents

Introduction.....	15
Section 1.0	
Software Version Explanation.....	15
Section 2.0	
Safety & Hazard Information	16
2.1 Text and Character Representation.....	16
2.2 Graphical Representation	17
Section 3.0	
Publication Conventions	17
3.1 General Paragraph and Character Styles.....	17
3.2 Page Layout.....	18
3.3 Computer Screen Output/Input Character Styles	19
3.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys).....	19
 Chapter 1	
LFC Procedures.....	21
Section 1.0	
Scope.....	21
Section 2.0	
Preliminary Requirements.....	21
2.1 Tools and Test Equipment.....	21
2.1.1 Software Requirements	21
2.1.2 Software Versions.....	21
2.2 Required Conditions	22
2.2.1 Hardware-Related Prerequisites.....	22
2.2.2 Software Load Prerequisites.....	22
Section 3.0	
Procedure.....	22
3.1 General Information	22
3.1.1 LightSpeed Plus GRE SW Load Checklist	22
3.1.2 Passwords	23
3.1.3 Other General Information	23
3.2 Save System State	23
3.3 Load from Cold Installation Procedures.....	24
3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer	24
3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer....	26
3.3.3 LFC — CDROM OS Installation on DARC	31
3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer.....	33
3.4 Software Patch Installation & Uninstallation Procedure.....	34
3.4.1 How to install	34
3.4.2 How to uninstall	35
3.5 Reboot the System	35

3.6	Start Up the CT Applications	35
3.7	Configure User's System Access	36
3.8	HIPAA Present set to ON — Admin Screen	38
3.9	Restore System State	38
3.10	Retro Recon Test	39
3.11	FLASH Download	39
3.12	Install Options	40
3.13	Set Time and Date	41
3.14	Restart the System	42
	3.14.1 Restarting the System	42
	3.14.2 If HIPAA Present set to ON — Admin Screen	42
3.15	LightSpeed Plus GRE Console Information Sheet	43
3.16	DVD Interchange Software Kit installation	43
3.17	Class C Software Installation	44
	3.17.1 Enabling Proprietary Service Software	44
	3.17.2 Removing Class C Software	45
3.18	Class M (InSite) Software Installation	45
	3.18.1 Load Class M Software	45
	3.18.2 Removing Class M Software	46

Appendix A

System State Conversion 47

Section 1.0

Introduction 47

Section 2.0

Procedure 47

Appendix B

Example config/INFO Output 49

Section 1.0

Sample Output 49

Appendix C

DARC OS Load Pop-Up Boxes 51

Section 1.0

Introduction 51

Section 2.0

Procedure 52

2.1	Error - Cannot connect with DARC	52
2.2	Error - BMC connection on DARC node is not responding	53
2.3	Error - Cannot send boot command to DARC	54
2.4	SOL Service - Run the Serial Over LAN Service CD	55
2.5	Error - Cannot send a complete signal to DARC	56
2.6	Information - DARC Node loaded OS but complete signal not sent	57
2.7	DARC Node - Cable Checkout	59

2.8	DARC Node - Remote Shell	59
2.9	DARC Node --- Serial Over LAN Checkout	60

Appendix D

General Information	61
----------------------------------	-----------

Section 1.0

Introduction	61
---------------------------	-----------

Section 2.0

Procedures.....	61
------------------------	-----------

2.1	Remote Shell — Check Internal Network	61
2.2	HP Computer CDROM — Mount and Unmount Process	61
2.3	Peripheral Tower DVD — Mount and Unmount Process.....	62
2.4	Peripheral Tower DVD — Initializing a DVD	62
2.5	Peripheral Tower MOD — Mount and Unmount Process.....	62

Introduction

Purpose: This introduction section identifies the software version explanation and publication conventions used in this document. The conventions information helps the reader identify safety and general material that is considered important by it's format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task.

Please become familiar with the conventions used within this publication before proceeding.

Section 1.0 Software Version Explanation

This section identifies the CT products and software versions included in this document. After installation you can check the product name and software version on your system, do the following:

- 1.) Open a shell.
- 2.) Type: **check_config** ENTER.
- 3.) The "check_config" command displays a number of output lines, including "Product" and "sw_version" (software version). Refer to the [Table A](#) below.

SYSTEM	MODEL	"PRODUCT"	"SW_VERSION"
LightSpeed 5.X Plus	--	LightSpeed Plus	04BW29.02.HP.P.M4.G

Table A Software Compatibility

Note: The GRE Common Applications software supports multiple hardware configurations via a single software set. The new software version string has the following format:

`<software_label>.<G>_<T>_<D>_<R>`

The "software label" format is: YYLWxx.ZZ

Where: YY = Year. For example 04.

L = Location. M - Milwaukee, H - Hino, B - Beijing.

Wxx = Fiscal Week. For example, w15.

ZZ = Revision number for each build/roll.

Refer to [Table B](#) for an explanation of the `<G>_<T>_<D>_<R>` format.

G = GANTRY TYPE	T = TUBE TYPE	D = DAS TYPE	R = RECON HW TYPE
H1 = LightSpeed 1.X	P = Performix	S4 = SDAS4	P = Pegasus
H2 = LightSpeed 2.X, 3.X, 4.X	O = OEM	M4 = MDAS4	G = GRE
HP = LightSpeed 5.X	H = Hercules	M8 = MDAS8	
WB = LightSpeed 5.x RT		M16 = MDAS16	
QX = HighSpeed QXi		G16 = GDAS16	

Table B Common Applications Software Naming Conventions

The variables G, T, D, and R are generated dynamically based on the type of hardware configuration this software is installed on. The information for the dynamic part of the string is obtained from the INFO files which are located in /usr/g/config directory.

An example string based on the above format is as follows:

04MW13.7.H2_P_M4_G

The break down is as follows: This is 2004 software release at Milwaukee in Fiscal Week 13 with a revision number "7". The hardware configuration is "H2" gantry with "Performix" tube, "MDAS4", and "GRE" recon hardware.

Section 2.0 Safety & Hazard Information

2.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that is applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential Hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.

CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists which can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects

DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS WHICH WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION

WARNING
ROTATING
EQUIPMENT
BARE WIRES

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.

NOTICE
Equipment
Damage
Possible

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube

- Coasting the rotor through resonance.
- It's important that the reader not ignore hazard statements in this document.

2.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 3.0 Publication Conventions

3.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles and shading are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

Purpose: Introduces and provides meaning as to the information contained within the

chapter, section or subsection, such as used at the beginning this chapter for example.

Note: Conveys information that should be considered important to the reader.

Example: Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.

Comment: Represents “additional” information that may or may not be relevant.

3.2 Page Layout

Publication Part Number & Revision Number

The current section and its title are always shown in the footer of the left (even) page.

An exclamation point in a triangle is used to indicate important information to the user.

Paragraphs preceded by **Alphanumeric** characters (e.g. numbers) contain information that must be followed in a **specific order**.

Publication Title

The current chapter and its title are always shown in the footer of the right (odd) page.

Paragraphs preceded by a **symbol** (e.g. bullets) contain information that has **no specific order**.

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd numbered page footers indicate the current chapter, its title and current page number. Even page footers show the current section and its title, as well current page number.

3.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicated direction. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example:
Fixed Output

This paragraph denotes computer screen fixed output. It's output is fixed from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths and text.

Example: Variable Output	<i>This paragraph denotes computer screen output that is variable. Its output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput></i>
-----------------------------	---

Example: Fixed Input	This paragraph denotes fixed input. It's typed input that will not vary from application to application. Fixed text the user is required to supply as input.
-------------------------	---

Example: Variable Input	<i>This paragraph denotes computer input that can vary from application to application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.</i>
----------------------------	--

3.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: Hard Keys	A power switch <u>ON/OFF</u> or a keyboard key like <u>ENTER</u> is indicated by applying a character style that uses both over and under-lined bold text that is bold. This is a hard key.
-----------------------	---

Example: Soft Keys	Whereas the computer <u>MENU</u> button that you would click with your mouse or touch with your hand uses over and under-lined regular text. This is a soft key.
-----------------------	--

Chapter 1

LFC Procedures

Section 1.0

Scope

This LFC (Load from Cold) procedure is written specifically for the LightSpeed Plus GRE system and should be used ONLY when loading the applicable software package.

Section 2.0

Preliminary Requirements

2.1 Tools and Test Equipment

2.1.1 Software Requirements

The LFC requires a total of 4 CDRoms. For software versions specific to each installation, see [Section 2.1.2](#).

- 1.) Console Operating System (OS) software will be loaded on both the HP Host Computer and the DARC Node,
- 2.) Specific LightSpeed Applications software will be loaded on the HP Host Computer, and
- 3.) Lightspeed DARC Applications software will be loaded on the HP Host Computer.

2.1.2 Software Versions

Current versions of software for LFC on **LightSpeed Plus**:

5111742 – LightSpeed Plus Xtreme Software Set

- **5114635 – LightSpeed / Linux Console Operating System**
Version – GEMS Linux 1.7.10
- **5117625 – LightSpeed Plus Applications Software (disk 1)**
Version – 04BW29.02.HP.P.M4.G
- **5117627 – LightSpeed Plus Applications Software (disk 2)**
Version – 04BW29.02.HP.P.M4.G
- **5114636 – LightSpeed DARC Applications**
Version – 04BW29.02.HP.P.M4.G

2.2 Required Conditions

2.2.1 Hardware-Related Prerequisites

- 1.) Verify the Peripheral Tower DVD RAM power is applied.
- 2.) Verify the Peripheral Tower terminator LED is lit (located at the rear).
- 3.) Verify the HP Host Computer DVD ROM drive does not have a CD in it.
- 4.) Verify and record specific system hardware configuration.

- a.) Open a shell.
- b.) Type: `cat /usr/g/config/INFO` ENTER
- c.) Record screen information in [Section 3.15](#). For an example output, see [Appendix B](#).

2.2.2 Software Load Prerequisites

- 1.) With Applications up, select the SERVICE icon to access the Netscape: Service Desktop.
- 2.) Select UTILITIES, SYSTEM STATE with Applications up. Save Protocols if applicable.
- 3.) Record Autovoice Volume control settings (ALT-F3 by Toolchest, upper right corner).

Section 3.0 Procedure

3.1 General Information

3.1.1 LightSpeed Plus GRE SW Load Checklist

- ☐ Save System State (be sure save any customer reformat files that might exist).
- ☐ Verify and record specific system hardware configuration.
Open a shell and type: `cat /usr/g/config/INFO` ENTER
- ☐ Install the Console Operating System (OS) on the HP Host Computer (GEMS2 or iGEMS2).
- ☐ Select OK by pressing the ENTER key and immediately remove the OS CD.
- ☐ Install the LightSpeed Applications SW on the HP Host Computer – Yes to autorun.
- ☐ Select LOAD and YES to InfoFile (restore the System State) then, select ACCEPT.
- ☐ Select YES to reboot and remove LightSpeed Applications CD from HP Host PC.
- ☐ Install the OS on the DARC Node, then type as required.
- ☐ Remove OS CDROM from DARC Node.
- ☐ Install the DARC Applications SW on the HP Host PC, type as required.
- ☐ Remove the DARC Applications SW from the HP Host PC.
- ☐ Install service software as applicable; reboot.
- ☐ Select CANCEL to stop Applications Software from coming up or OK as required.
- ☐ Open a shell and type: `st` ENTER.
- ☐ Perform the Retro Recon Sanity Check.
- ☐ Restore System State.
- ☐ Perform the Flash Download Update.
- ☐ Install the Options.
- ☐ Reboot the system.
- ☐ Perform Scout, Helical, and Axial Sanity Scans.

3.1.2 Passwords

- root: `#bigguy`
- ctuser: `4$app`

3.1.3 Other General Information

Refer to [Appendix D](#) for the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

3.2 Save System State

- 1.) Insert the DVD into the Peripheral Tower DVD RAM tower drive.
- 2.) Select: SERVICE DESKTOP.
- 3.) Select: UTILITIES.
- 4.) Select: SYSTEM STATE.
- 5.) Click ALL to select all the cals, characterizations, etc.
- 6.) Click SAVE.
- 7.) If applicable, click OK when the following message appears: System State Media Status: Please insert a DVD or MOD into the drive and press Save again.
- 8.) When completed select DISMISS.
- 9.) Save all reformat / recon protocols.
- 10.) Close the Service Desktop window in the upper left corner of the screen.
- 11.) Sites performing the next sub-section should not remove the DVD from the Peripheral Tower DVD RAM drive.

3.3 Load from Cold Installation Procedures

Use these procedures to perform a LFC from a grouping of CDROM's.

3.3.1 LFC — CDROM Operating System (OS) Installation on HP Host Computer

Note: System State SAVE must be performed before shutting down apps (Application Software).

- 1.) Verify the System State was saved.
- 2.) Remove the GRE Operator Console's front cover.
- 3.) Insert the **LightSpeed / HiSpeed Linux Console Operating System** CD-ROM into the DVD ROM drive on the HP Host Computer.

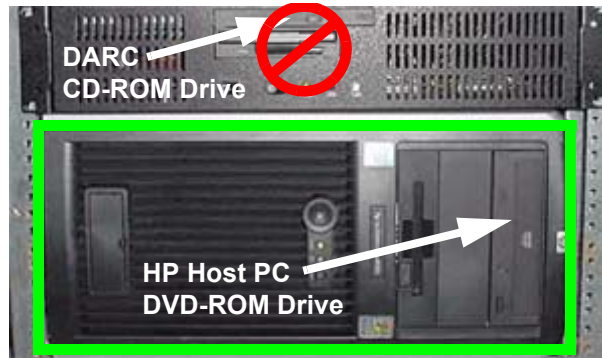


Figure 1-1 HP Host PC DVD-ROM Drive Location

- 4.) If Applications are up, shut down the machine by clicking the red SHUTDOWN icon in the upper left corner of the screen –or– Power OFF the console.



Figure 1-2 Shutdown Icon

The following options appear:

Attention:
Shutdown the system?
Logout User
Restart
Shutdown

- 5.) Power ON the Console or Select: RESTART then OK to shut down and restart the system.
As the computer restarts, the booting process messages appear.
After the booting process completes, the boot : prompt appears.
- 6.) At the boot prompt, type the following depending on system type:
Type: **GEMS2** ENTER (for English Keyboard only)
or: **iGEMS2** ENTER (international command only)

Note: The Console Operating System load takes approximately 10 minutes. If the boot : prompt does not appear, then either the CDROM is bad or the Console Operating System CDROM is not installed in the drive. Typing **GEMS** will result in both scan and display being on one monitor and require a LFC.

-
- Note: Failure to remove the LightSpeed Console Operating System CDROM results in an OS reload.
- 7.) After the OS is loaded, a red button appears and displays OK. Press ENTER, and then immediately remove the OS CD (within 10 seconds) when it ejects.
 - 8.) The HP Host Computer reboots. **Do not insert the LightSpeed Plus Applications Software CDROM** into the HP Host Computer until the system reboots and the `[root@localhost]` window appears, displaying the `[root@localhost root]#` prompt.

3.3.2 LFC — CDROM Application Software Installation on the HP Host Computer

Two separate Application Software packages (for the HP Host Computer and the DARC Node) are loaded via CDROMs onto the HP Host Computer.

WARNING FAILURE TO INSTALL THE CORRECT APPLICATION SOFTWARE AND SELECT THE CORRECT SYSTEM HARDWARE CONFIGURATION WILL RESULT IN SYSTEM DOWNTIME AND CIRCUIT BOARD REPLACEMENT.

Note: For platform-specific software versions, see [Section 2.1.2](#).

- 1.) After the system reboots and the window appears with the `[root@localhost root]#` prompt insert the **LightSpeed Plus Applications Software** CDROM into the DVD-ROM drive on the HP Host PC.

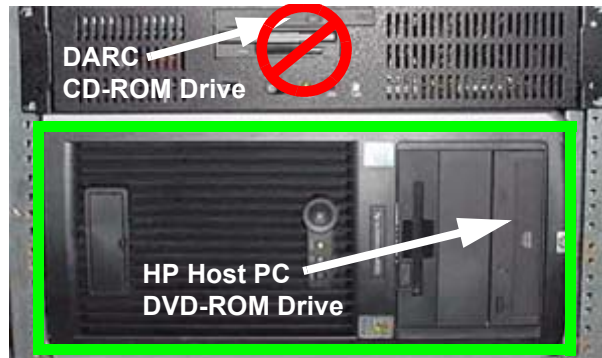


Figure 1-3 HP Host PC DVD-ROM Drive Location

- 2.) After approximately one minute a pop-up box appears, asking:

`Do you wish to run /mnt/cdrom1/autorun?`

Note: If the message does not appear, eject the Applications CD, re-insert it, and try again.

- 3.) Click YES. After approximately one minute, an Installation Utility window appears.

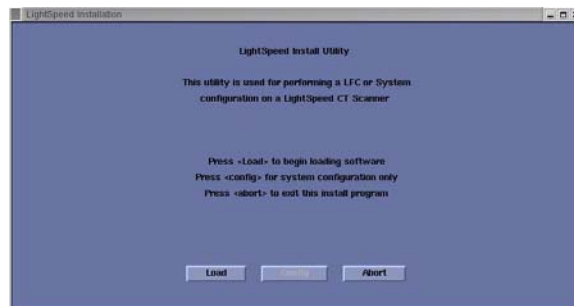


Figure 1-4 Lightspeed Install Utility window

- 4.) Click LOAD. A prompt message appears asking if you wish to load the INFO file from the System State DVD.
- 5.) If you have a System State DVD, select YES. When prompted, ensure the System State DVD is in the DVD drive in the Peripheral Tower (on top of the console) and then select OK. If you do not have a System State, select NO.

- 6.) Click the SYSTEM button.

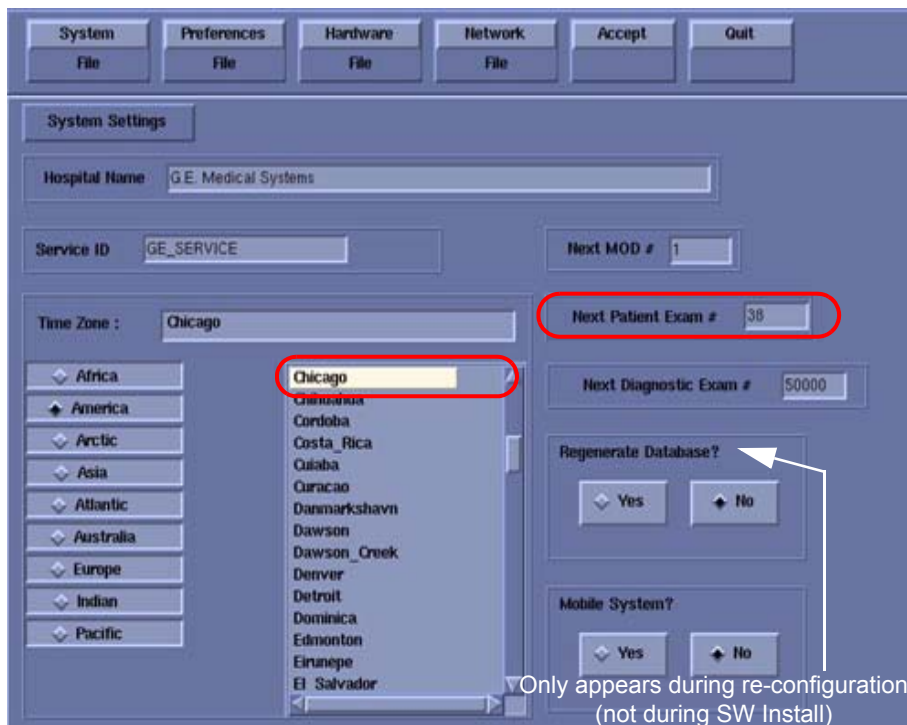


Figure 1-5 System Settings screen

- 7.) Verify that the **Next Patient Exam #** on the System Settings screen is either the value restored from Save System State, else = 1.
- 8.) Verify that the proper **Time Zone** is selected (example: Chicago).
- 9.) Click the PREFERENCES button.

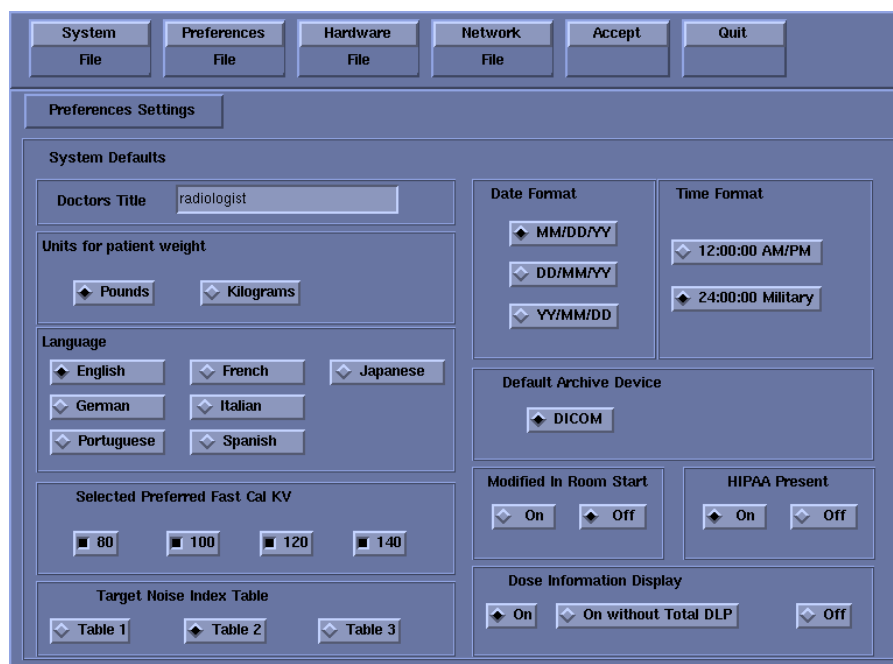


Figure 1-6 Preferences screen

- 10.) Select the units for **patient weight**: (POUNDS or KILOGRAMS).
 - 11.) Select **Language**: (ENGLISH or OTHER).
 - 12.) Select **Preferred FastCal KV** - select the kV to be calibrated during FastCal. (Default selections are 120 and 140). These kVs should include all kVs that the site uses for patient scanning.
 - 13.) Select **Target Noise Index Table**: default is TABLE 2.
 - 14.) Verify proper **Date Format** is selected (Example: MM/DD/YY).
 - 15.) Verify proper Time Format is selected (Example: 24:00:00 MILITARY).
 - 16.) Default **Archive Device**: add your device here (Example: DICOM.)
 - 17.) Verify **Modified in Room Start** is set to OFF.
 - 18.) Set **HIPAA Present** to OFF unless the customer requests HIPPA to be on.
 - 19.) Select the site preferred **Dose Information Display** option for the site to use in monitoring calculated Patient Dose:
 - a.) Select ON (full CTDIw Display)
 - b.) Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - c.) Select OFF (no CTDIw Display)
- Note: A scroll bar below this menu may be present. It contains the System Information from the INFOfile System State.
- 20.) Click the HARDWARE button. (The Hardware Settings screen is different for each system).

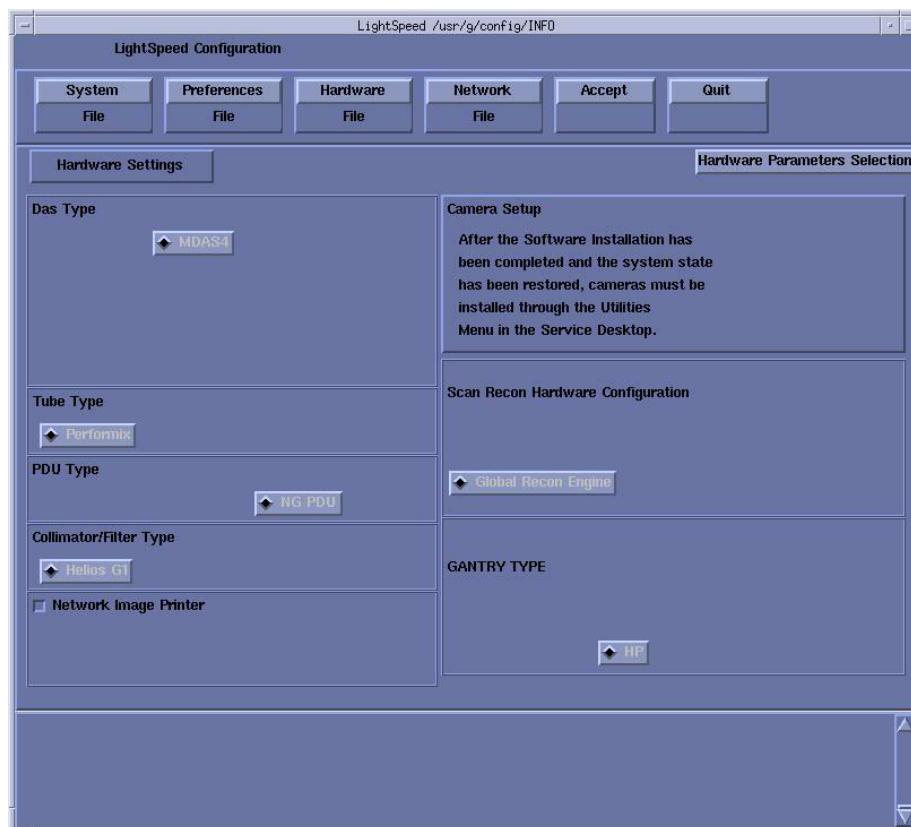


Figure 1-7 Hardware Settings screen

- 21.) Click on **Hardware Parameters Selection -> LightSpeed Plus w/ MDAS**.
- 22.) Refer to [Table 1-1](#) to verify: DAS Type, Tube Type, PDU Type, and Gantry Type. If not correct, click the HARDWARE PARAMETERS SELECTION button then select the correct system

configuration from the displayed list.

SYSTEM	DAS	TUBE	PDU	GANTRY
LightSpeed 1.X (QX/i)	SDAS4	Performix	CT COMPACT	H1
LightSpeed 2.X (Plus, QX/i)	SDAS4	Performix	CT COMPACT	H2
HiSpeed QX/i	MDAS4	Performix	CT COMPACT or NG PDU	HiSpeed QX/i
LightSpeed 3.X (Plus, QX/i)	MDAS4	Performix	CT COMPACT or NG PDU	H2
LightSpeed 3.X (Ultra)	MDAS8	Performix	CT COMPACT or NG PDU	H2
LightSpeed 4.X (H16)	MDAS16	Performix	CT COMPACT	H2
LightSpeed 5.X Pro ¹⁶ (100kW) (HP100)	GDAS16 or MDAS16	Hercules	NG PDU	HP
LightSpeed 5.X Pro ¹⁶ (80kW) (HP80)	GDAS16 or MDAS16	OEM	NG PDU	HP
LightSpeed 5.x RT	MDAS4	OEM	NG PDU	WB
LightSpeed 5.x Plus	MDAS4	Performix	NG PDU	HP

Table 1-1 DAS, Tube, PDU, and Gantry Type Selections

NOTICE Scanning with the wrong Tube Type will damage the tube! Before continuing, verify the Tube Type is correct per [Table 1-1](#).

23.) Verify Collimator/Filter Type: HELIOS G1.

24.) Verify Scan Recon Hardware Configuration: GLOBAL RECON ENGINE.

25.) Check the **Network Image Printer** selection **ONLY** if the system has an active network printer ("active" means the printer is installed, cabled, and powered ON). If the printer is not active, the software install will fail.

26.) Click the NETWORK button (site will be different).

Figure 1-8 Network Settings screen

- 27.) Select the **Suite Name**: add your name here (Example: *lib2*).
- Note: The Suite Name must start with a letter, followed by 3 alphanumeric characters Total must be four characters long. The name of the OC interface is <Suite Name>_oc within the scanner's subnet. It is suggested that you choose CT01 as its name, unless a different Suite Name is required.
- 28.) Select the Host Name: add your host name here (Example: *lib1ab2*).
- Note: The **Host Name** identifies the hostname and **AE Title** of the scanner. It:
- MUST NOT be <Suite Name>_oc or <Suite Name>_OC.
 - MUST NOT exceed 16 Characters.
 - MUST only contain the following characters: A thru Z, a thru z, 0 thru 9, - and _
- 29.) Set the **IP Address**: add your address here (Example: *3.57.255.55*).
- 30.) If the **IP Address** starts with **172.xx.xx.xx**, then check the **Change DARC Subnet?** box and enter the following value: *169.254.0*
- 31.) Set the **Netmask**: add your address here (Example: *255.255.252.0*).
- 32.) Set the **Broadcast Address**: add your address here (Example: *3.57.255.255*).
- 33.) Set the **Default Gateway**: add your address here (Example: *3.57.252.254*).
- Note: If the site does not have a Gateway, enter the value *1.1.1.1*. This field MUST NOT be blank!
- 34.) If there is an **Internal Option – Internal Subnet** setting displayed and checked, then **un-check it**. (This option might not appear on your system; it is not shown is [Figure 1-8](#).)
- 35.) Verify with customer's network administrator if they have NIS. If they do, ensure the ADVANCED OPTIONS and USE NIS boxes are checked. If they don't, ensure these selections are not checked.
- a.) Enter **Domain Name**: add your name here (get name from customer's network administrator) (Example: *CTSYSTEMS*.)
 - b.) Enter the **IP Address** of Internal Server: add your IP address here (get IP address from customer's network administrator) (Example: *3.70.56.18*).
- 36.) Click the **ACCEPT** button at the right corner of the User Interface.
- 37.) An Install INFO pop-up box is displayed. See [Figure 1-9](#) for a typical Install INFO message (the Install INFO message is different for each system configuration).

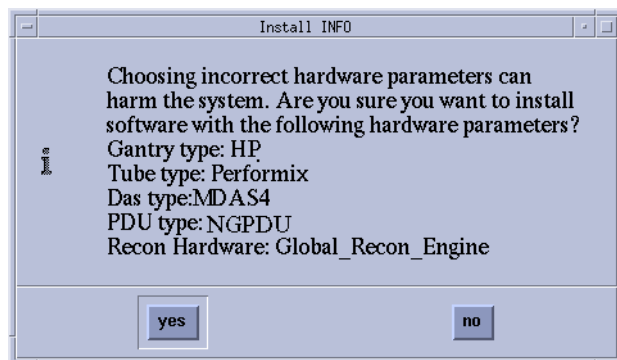


Figure 1-9 Hardware Confirmation Screen (Example - LightSpeed Plus SDAS Shown)

- 38.) If the information is correct, click the YES button.
- Note: If the information is not correct, click the NO button. This returns you to the Hardware screen. Click the HARDWARE PARAMETERS SELECTION button and select the correct system configuration from the displayed list. Click the **ACCEPT** button again, followed by the YES button on the Install INFO pop-up to accept the new hardware configuration.
- 39.) The following pop-up message appears: Please make sure the CT Application SW is in the drive - the system will be rebooted. Ignore this message.

- Note: Do not remove the **GRE/Xtream Applications Disk #2** from the HP Host Computer until instructed to do so.
- 40.) As the system reboots, the following Winterm window message appears: Starting LFW procedure (~13 minutes to load). A shell Installation screen appears and the application software starts loading.
 - 41.) When the system prompts you to reboot (after approximately 13 minutes), answer YES. The system is going down for reboot NOW.
 - 42.) After approximately 3 minutes, a pink pop up window appears, stating CT Software Auto-Start Disabled. In this box select: OK.
 - 43.) Press the button on the DVD ROM drive to remove the **GRE/Xtream Applications Disk #2** from the HP Host Computer.

3.3.3 LFC — CDROM OS Installation on DARC

The LightSpeed / HiSpeed Linux Console Operating System CD-ROM will be used again; however, this time it will be loaded onto the DARC Node. If any problems arise, verify the correct CD is installed.

- 1.) Open a Unix Shell.
- 2.) Type the following to verify hardware configuration:

```
ctuser: cat space /usr/g/config/INFO ENTER
```

Verify DAS type (MDAS4 or MDAS8), Tube type (Performix), PDU type (NG PDU), Gantry type (HP), and Scan Recon Hardware Configuration (GRE). If configuration is not correct, restart the software install (see [Section 3.3.1](#)).

- 3.) Type the following:

```
ctuser: su SPACE - ENTER
```

```
password: #bigguy ENTER
```

```
root: cd SPACE /usr/g/scripts ENTER
```

```
root: ./start_darcOS ENTER
```

An Attention window appears. See [Figure 1-10](#).

If the ./start_darcOS script does not work, verify the Global Recon Engine has been selected (see General Information).



Figure 1-10 Insert OS CD Pop-Up Window

- 4.) Follow the instructions in the Attention window and then insert the **LightSpeed Plus Linux Console Operating System** CD-ROM into the DARC Node drive.

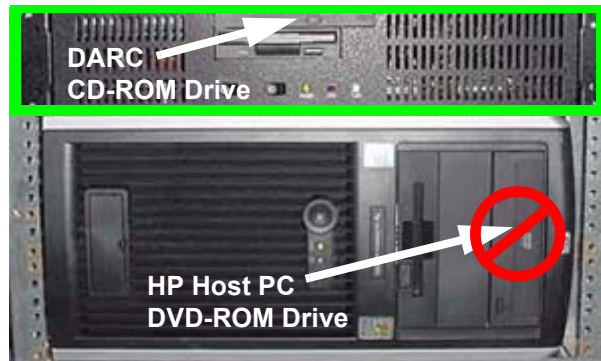


Figure 1-11 DARC CD-ROM Drive Location

- 5.) Click on **OK**.

A sh window displays initial information, repetitive `dpcli>` commands, then the message below:

The OS load on the DARC Node is started. This takes about 9-11 minutes.
##.## % done.

Note: The DARC OS Load will stop prematurely **ONLY** if an error is encountered. An Error Dialog Box will appear with a brief description of the error. Select OK after reading the instructions displayed on the Monitor. Also, if any problems arise, verify the correct CD is installed.

- 6.) Refer to the `%done` status displayed during the DARC OS load. When the DARC OS load has completed, a the following message is displayed.

The OS load is complete. Please remove the OS disk from the DARC disk drive. DARC is rebooting. Wait for the DARC to boot up..

- 7.) The OS disk CD ejects automatically. Remove the OS disk from the drive and close the drive.

Note: If the OS disk CD does not eject, then you may be instructed to eject it manually. If a message pop-up box does not appear and the DARC OS CD does not eject, then press the

DARC Node front panel RESET Switch (⏻). Manually eject the CD after approximately 2 minutes. See [Figure C-8](#) for location of DARC Reset Switch.

- 8.) The DARC OS 'Load Window' should disappear when the load has completed successfully. An information window should appear. See [Figure 1-12](#).

Note: Other pop-up windows may appear with instructions if there was a problem with or during the DARC OS load. Refer to [Appendix C](#) for additional information on these pop-up windows.

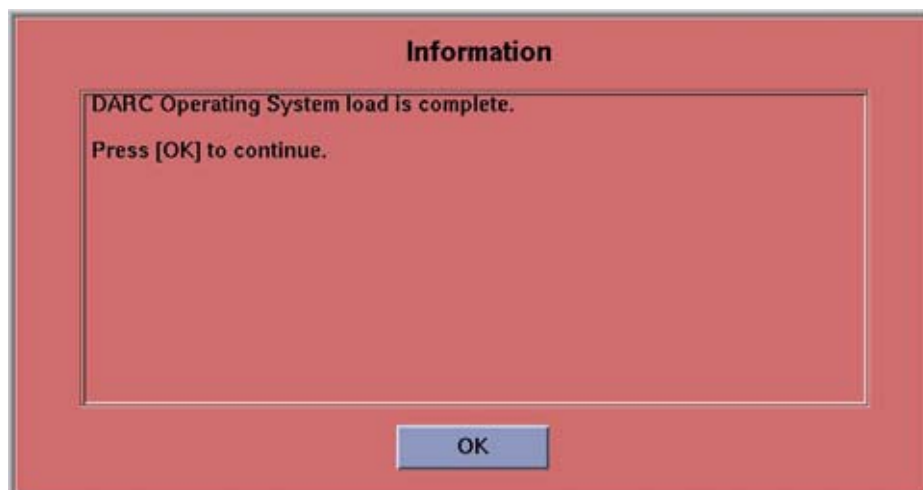


Figure 1-12 DARC OS Load Complete Pop-up Window

- 9.) Click on OK.
- 10.) In upper left select FILE and close the shell (or you may use this for the next sub-section).
- 11.) This completes the OS load on the DARC.
- 12.) Next: load the DARC Applications in the HP Host Computer. See [Section 3.3.4](#).

3.3.4 LFC — CDROM DARC Application Installation on HP Host Computer

This section first verifies the DARC OS was successfully Loaded. The DARC Applications Software CD-ROM will then be used and loaded onto the HP Host Computer here. If any problems arise, verify that the correct CD is installed.

- 1.) Insert the **LightSpeed Plus DARC Applications** CDROM into the DVD-ROM drive on the HP Host Computer.

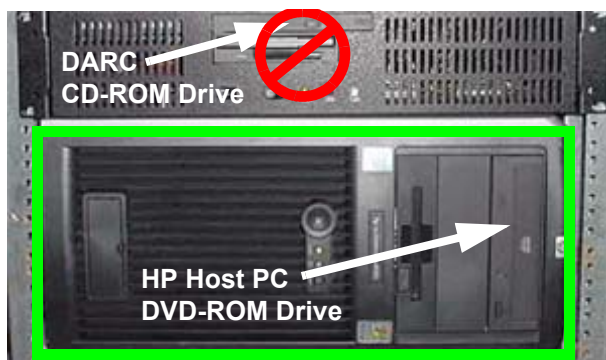


Figure 1-13 HP Host PC DVD-ROM Drive Location

- 2.) Open a Unix Shell and type the following:
ctuser: su SPACE - ENTER
password: #bigguy ENTER
- 3.) Verify the DARC OS load was successful by logging into the DARC Node:
root: rsh SPACE darc ENTER
[root@localhost root] (If the prompt is displayed exactly as shown, then the login was successful and the DARC OS was loaded properly.)
[root@localhost root] exit ENTER
- 4.) If login was successful, continue with DARC Apps load:
root: cd SPACE / ENTER
root: mount SPACE /mnt/cdrom ENTER
root: cd SPACE /mnt/cdrom ENTER
root: ./copy_rpms ENTER
The following messages appear.
Copying darc rpms...
Please wait...
Done...
- 5.) Type the following:
root: cd SPACE / ENTER
root: umount SPACE /mnt/cdrom ENTER
root: eject ENTER
- 6.) Remove the LightSpeed Plus DARC Applications CDROM from the HP Host Computer DVD-ROM drive.

7.) Type:

```
root: cd SPACE /usr/g/scripts ENTER
```

```
root: ./start_darc_load ENTER
```

A shell window pops up and the load begins. **A successful load takes 4–5 minutes.**

Continue the procedure when the load window goes away.

Note: If the time to load the DARC apps is less than 1 minute (~30 seconds) then the copy rpms command was not entered successfully.

8.) Close the shell (do not use for the next process).

3.4 Software Patch Installation & Uninstallation Procedure

3.4.1 How to install

1.) Inset the CD-ROM into the DVD drive in the Tower.

2.) Application shutdown.

NOTE: Make sure the application has been shut down.

3.) Select **Unix shell** from desktop Menu located on the upper right of the screen. Then type the following responses to the indicated prompts.

```
Type: su ENTER.
```

```
Type the root password: #bigguy and press ENTER.
```

```
Type: mount /dev/dvd /mnt/cdrom ENTER.
```

```
Type: cd /mnt/cdrom/bin ENTER.
```

```
Type: ./installPackage ENTER.
```

4.) Then **"Do you install patch software?"** window appears. Click on **YES**.

5.) Finally **"Installation successful"** will display.

6.) Type: cd / ENTER.

```
Type: umount /mnt/cdrom ENTER.
```

```
Type: exit ENTER.
```

7.) Close the shell.

NOTICE When perform uninstallation, be sure to use the same software patch as installed.

3.4.2 How to uninstall

1.) Inset the CD-ROM into the DVD drive in the Tower.

2.) Application shutdown.

NOTE: Make sure the application has been shut down.

3.) Select **Unix shell** from desktop Menu located on the upper right of the screen. Then type the following responses to the indicated prompts.

```
Type: su ENTER.
```

```
Type the root password: #bigguy and press ENTER.
```

```
Type: mount /dev/dvd /mnt/cdrom ENTER.
```

```
Type: cd /mnt/cdrom/bin ENTER.
```

```
Type: ./uninstall_patch ENTER.
```

4.) Then **"Do you remove patch software?"** window appears. Click on **YES**.

5.) Finally **"Uninstallation succeed"** will display.

6.) Type: cd / ENTER.

```
Type: umount /mnt/cdrom ENTER.
```

```
Type: exit ENTER.
```

- 7.) Close the shell.

3.5 Reboot the System

- 1.) Open a Unix Shell.
- 2.) Type: `su` SPACE - ENTER
- 3.) Enter the root password: `#bigguy` ENTER
- 4.) Type: `sync` ENTER
- 5.) Type: `sync` ENTER
- 6.) Type: `reboot` ENTER
A message appears: The system is going down for reboot NOW!.
- 7.) After approximately 4–5 minutes a pink pop up window appears with the message:
CT Software Auto-Start Disabled.
Click OK.

3.6 Start Up the CT Applications

- 1.) When the Host Computer is up, open a Unix shell (window).
- 2.) Type: `st` ENTER or `startup` ENTER.
- 3.) A pink pop-up Attention Message appears: OC initializing. Please wait....
Read the Note(s) below.
- 4.) Click OK as needed to answer message prompts.
- 5.) If any Recon Selftest Failures are encountered, review the error log.

- Note:
- After the INITIAL Load From Cold, the IG needs to be powered on. This is only if the IG has never been involved in a LFC. The IG power is normally on later in the Host Computer initializing message and before the IG coming up message box. The IG Node is turned on during the RESETTING message (displayed on the right Display Monitor message box located below the Service and Shutdown Icon).
 - If a message concerning incorrect DAS configuration is encountered, review the Error Log for the Card List issue. Select from the following two methods to alleviate this issue: Flash Download Update or Power OFF the Axial Drive and HVDC at the Gantry service panel - turn OFF DAS Power for 1 minute - turn DAS Power back ON - turn Axial Drive and HVDC at the Gantry service panel back ON - Flash Download Update - verify error message has disappeared. If error message reappears then Troubleshoot.

3.7 Configure User's System Access

ATTENTION: The HIPAA has been disabled when the software was loaded on this system. Do not turn it on without the consent of your customer. Perform the following steps ONLY IF the customer has specifically requested for it to be enabled.

The list of users who will have access to the system must be added before that person can gain access to the system. Configuration of these users may be added by any user that has admin permission. The service or admin users are both able to add users. The admin users will typically be assigned by the customer administration and will add accounts for the system.

In order to add users, you will need to access the admin screen. To do this:

- 1.) Press the pink SHUTDOWN button.
- 2.) Select LOG OUT USER and then OK, to bring up the login/admin screen.

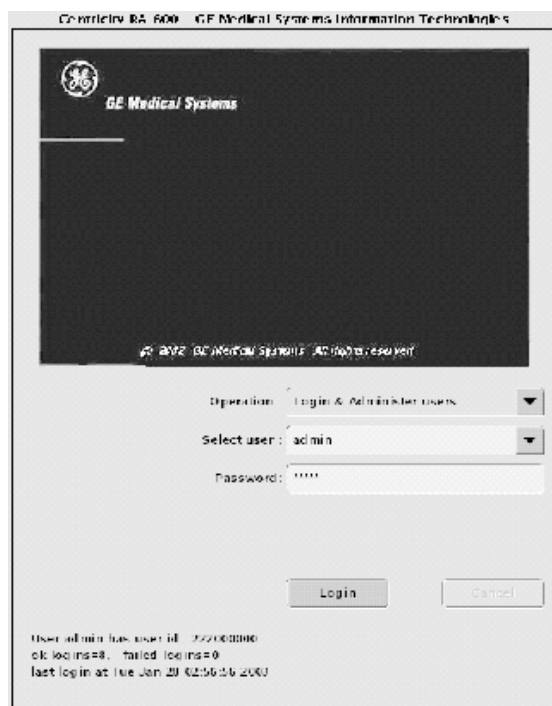


Figure 1-14 Login/Admin screen

- 3.) Select from the pull-down menu **Operation: Login and Administer Users**.
- 4.) Enter for the Username: **service** and Password: **4rhelp**, then select LOGIN.
- 5.) On the left hand of the screens you see tabs for admin functions. Select the USER, GROUPS PERMISSIONS tab.

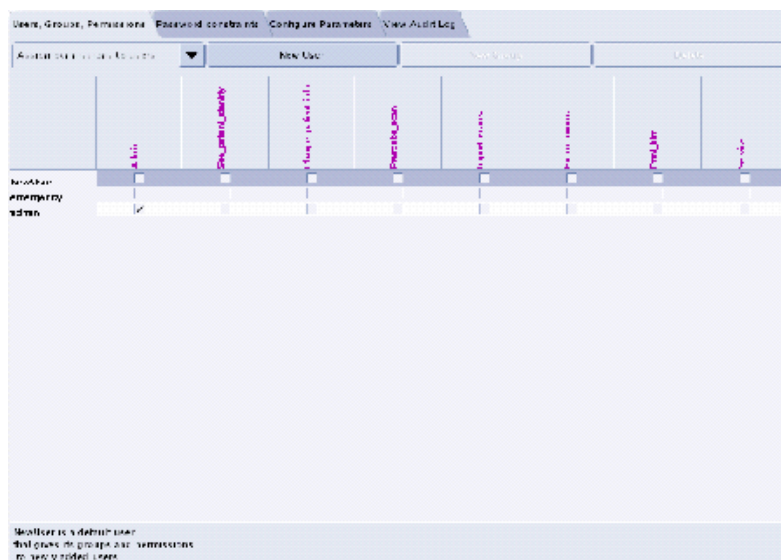


Figure 1-15 User, Groups, Permissions screen

- 6.) On the left hand of the screens you see tabs for admin functions. Select NEW USER button, which will pop up an entry to type a new user. Enter the user name and select OK.

Note:

- The user name must not contain spaces.
- The initial password will be the user name.
- The new user will be prompted to change the password the first time they log in.



Figure 1-17 Login/Admin screen

3.9 Restore System State

This procedure restores Characterization Data, Calibration Data, Protocols, etc., from your System State DVD. If you don't have a System State DVD, copy a similar DAS type bay's System State: ALL Characterizations and Calibrations in order to make the system operational.

- 1.) Insert the Save System State DVD into DVD RAM Peripheral Tower drive.
- 2.) Select: SERVICE DESKTOP.
- 3.) Select: UTILITIES.
- 4.) Select: SYSTEM STATE.
- 5.) Click the ALL selection which highlights the cals, characterization, etc.
- 6.) Make certain to click RESTORE.
- 7.) Verify the overall system state content has restored correctly (look for message "Save/Restore System State Completed Successfully"), then select CANCEL.
- 8.) When completed, select DISMISS or close the window.

Comment: If the Service Key is installed, press the (1), (2), (3) keys in order to proceed.

Note: If Scan Hardware Reset question appears – answer NO. This reset is performed during FLASH Download Update.

3.10 Retro Recon Test

The Reconstruction (not Scan) portion of the system can be tested at this point.

- 1.) Click on the RETRO RECON selection on the left Monitor.
- 2.) Click the rat gold series.
- 3.) Click SELECT SERIES.
- 4.) Click on CONFIRM.
- 5.) Verify the image(s) reconstruct and are displayed.
- 6.) Click on QUIT.

3.11 FLASH Download

- 1.) Select the SERVICE DESKTOP.
- 2.) Select in sequence: UTILITIES > FLASH DOWNLOAD.
- 3.) Click UPDATE. Ignore any initial errors.
- 4.) When the Flash Download process is complete, select DISMISS or Close the window.

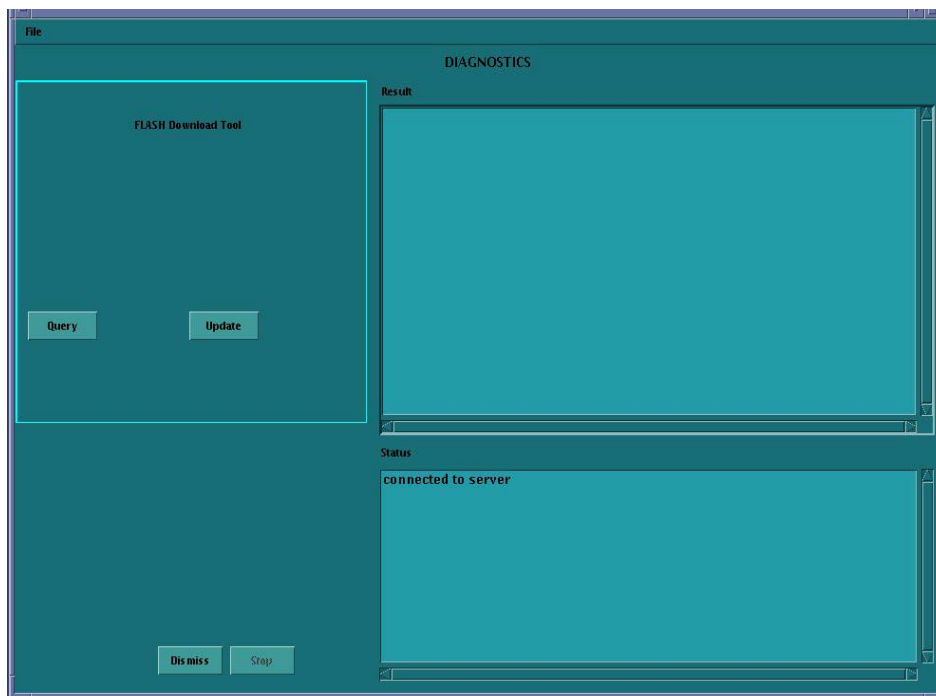


Figure 1-18 Diagnostics screen

- 5.) If there are any issues with the install, verify the reset switch at the gantry is not flashing and reset the 120 VAC, if necessary. Also, try to ping the following (CTRL-C and close the shell when finished pinging):
`ping SPACE orp ENTER`
`ping SPACE stc ENTER -OR- ping SPACE tgp ENTER`

3.12 Install Options

- 1.) Insert the Options DVD in the Peripheral Tower DVD RAM drive.
- 2.) With the Applications up, select the SERVICE DESKTOP.
- 3.) Select CONFIGURATION.

4.) Select INSTALL OPTIONS. A Software Options screen ([Figure 1-19](#)) appears.

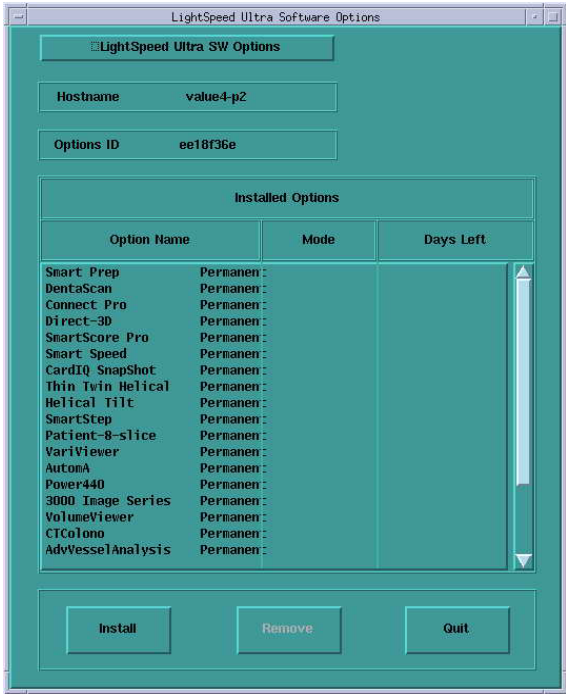


Figure 1-19 Install Options screen

5.) Click INSTALL. A Select Mechanism ([Figure 1-20](#)) window opens. Select the mechanism through which you want to install Option Keys.



Figure 1-20 Select Mechanism window

6.) Click PERMANENT. A Select Device ([Figure 1-21](#)) window opens. Select the mechanism through which you want to install the Permanent Option Keys.

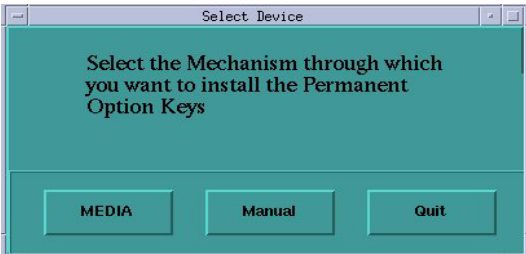


Figure 1-21 Select Device window

7.) Click the MEDIA button and insert the Options DVD or MOD.

Note: The Select Device illustration in [Figure 1-21](#) is outdated — the new window displays selections for MEDIA, Manual, and Quit. Screen requires an update.

- 8.) Click OK. Available options appear on the Software Options screen (Figure 1-22).

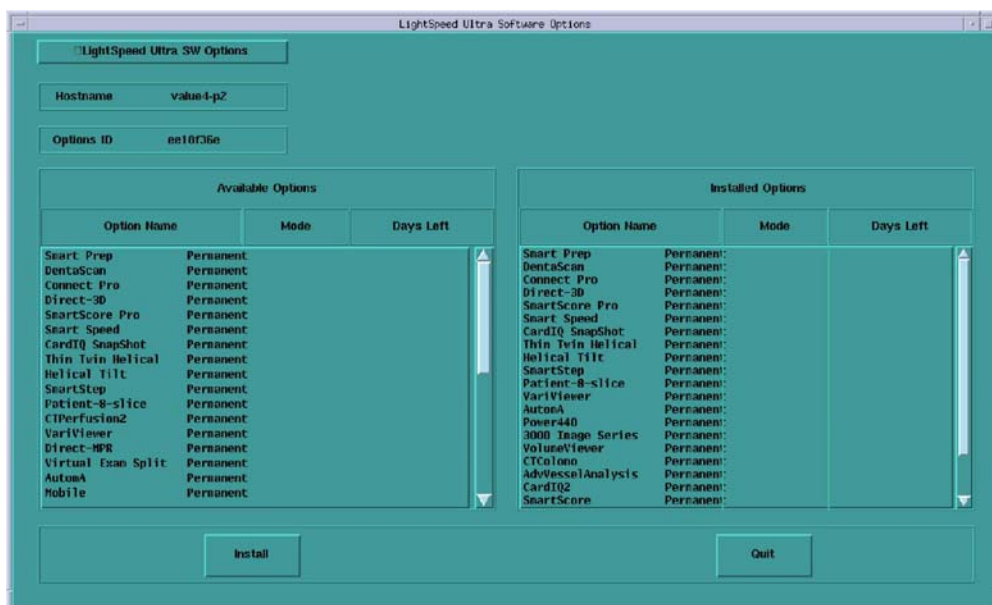


Figure 1-22 Software Options screen

- 9.) Pick options one at a time from the **Available Options** list and click on INSTALL to update each selection and place it on the Installed Options List.
- 10.) When the process is complete, click QUIT then QUIT to close the window.

3.13 Set Time and Date

Note: If the time and date is already correct, skip this procedure. Otherwise, you must set the date and time on the Host Computer.

With Application Software down, enter the time and date if needed.

- 1.) If Applications are up, select the SERVICE DESKTOP icon, UTILITIES, then APPLICATION SHUTDOWN.
- 2.) Open a Unix shell.
Type: `su` SPACE - ENTER.
Type the root password: `#bigguy` and press ENTER
- 3.) Type: `setdate mmddhhmmYYYY` (e.g.: `setdate 050409002003`) and press ENTER.

Note: Alternate method exists

You may also type: `setdate` ENTER and you will be prompted through the individual entries. Where:

`mm` is month (01-12)

`dd` is day (01-31)

`hh` is hour (00-23)

`mm` is minutes (00-59)

`YYYY` is year (1980-2030)

- 4.) The `setdate` program will set and display the time and date for on the Host Computer. Verify the date and time on the Host Computer is correct.
- 5.) To exit the root session, Type: `exit` ENTER.

- 6.) Slide the computer tray back into position, properly secure it, and replace the console's front cover.

3.14 Restart the System

3.14.1 Restarting the System

To restart the system:

- 1.) Open a Unix shell and type: `reboot` ENTER
- 2.) Select OK as needed to answer messages. Note any Recon Failures.
- 3.) Select the red SHUTDOWN ICON. A set of options appears.



Figure 1-23 Shutdown Icon

- 4.) Select: RESTART then OK to shutdown and Restart the system.

3.14.2 If HIPAA Present set to ON — Admin Screen

When the system comes up the following screen will be displayed if HIPAA Present was set to On during the LOAD process. Verify the admin screen below has the following 'Select user':

- Select user: `admin`
- Password: `ctAdmin`

For a more thorough discussion of this topic, see [Section 3.8](#).

3.15 LightSpeed Plus GRE Console Information Sheet

- ☐ Site ID #: _____ Console Serial Number _____
- ☐ DAS Type: MDAS4 or MDAS8
- ☐ Tube Type: Performix
- ☐ PDU Type: NG PDU
- ☐ KV Values: 80 100 120 140
- ☐ Collimator/Filter Type: Helios G1
- ☐ Scan Recon Hardware Configuration: GLOBAL RECON ENGINE
- ☐ GANTRY TYPE: GANTRY HP
- ☐ Target Noise Index Table: Table 2
- ☐ Dose Info Display: On
- ☐ Modified in Room Start: Off
- ☐ HIPAA Present: Off
- ☐ Default Archive Device: DICOM
- ☐ Suite Name: _____
- ☐ Host Name: _____
- ☐ IP Address: ____ . ____ . ____ . ____
- ☐ Netmask: ____ . ____ . ____ . ____
- ☐ Broadcast Address: ____ . ____ . ____ . ____
- ☐ Default Gateway: ____ . ____ . ____ . ____
- ☐ Printer: _____
- ☐ Printer IP: ____ . ____ . ____ . ____
- ☐ Internal Option – Internal Subnet: "this item may not appear" do not modify contents
- ☐ Advanced Option: this item is checked
- ☐ Use NIS? Enter Domain Name: _____
IP Address of Internal Server: ____ . ____ . ____ . ____
- ☐ ConnectPRO Option Info: *CT## WLSNorm 4008 3.57.255.##*
- ☐ Installed Hpower ME6 PATCH A

3.16 DVD Interchange Software Kit installation

Prerequisite:

- CD-ROM Disk: DVD Interchange Software Kit V/R2.1

Procedure:

- 1.) Check Software Version
 - a.) Select **Unix shell** from desktop Menu located on the upper right of the screen. Then type the following responses to the indicated prompts.
Type: **`rpm -qa | grep dvd`** **ENTER**.
 - b.) If displayed list includes **dvd_interchange_2_1**, current version of the DVD Interchange Software is the latest. No installation is needed. Close the shell.
 - c.) If displayed list includes V/R2.0 or earlier, uninstall the DVD Interchange Software by referring to "How to uninstall" of the respective version (the same DVD Interchange Software Version as installed).

- 2.) How to install
 - a.) Insert the CD-ROM into the DVD drive in the Tower.
 - b.) Application shutdown.

Note: Make sure the application has been shut down.

- c.) Select **Unix shell** from desktop Menu located on the upper right of the screen. Then type the following responses to the indicated prompts.
Type: **su -** ENTER.
Type the root password: **#bigguy** and press ENTER.
Type: **mount /dev/dvd /mnt/cdrom** ENTER.
Type: **mkdir /usr/g/ctuser/DI** ENTER.
Type: **cp /mnt/cdrom/* /usr/g/ctuser/DI** ENTER.
Type: **cd /** ENTER.
Type: **umount /mnt/cdrom** ENTER.
Type: **cd /usr/g/ctuser/DI** ENTER.
Type: **./install -type hpower** ENTER.
 - d.) Finally "**installation process succeed**" will display.
 - e.) Type: **exit** ENTER.
 - f.) Close the shell.

NOTICE When perform uninstallation, be sure to use the same DVD Interchange Software Version as installed.

- 3.) How to uninstall
 - a.) Insert the CD-ROM into the DVD drive in the Tower.
 - b.) Application shutdown.

Note: Make sure the application has been shut down.

- c.) Select **Unix Shell** from desktop Menu located on the upper right of the screen. Then type the following responses to the indicated prompts.
Type: **su -** ENTER.
Type the root password: **#bigguy** and press ENTER.
Type: **cd /usr/g/ctuser/DI** ENTER.
Type: **./install -u -type hpower** ENTER.
 - d.) Finally "**uninstallation succeed**" will display.
 - e.) Type: **cd..** ENTER.
Type: **mv DI DI.lfc** ENTER.
 - f.) Type: **exit** ENTER.
 - g.) Close the shell.

3.17 Class C Software Installation

3.17.1 Enabling Proprietary Service Software

- 1.) Load Class C Software
 - a.) On the Operator Console, open a Unix Shell window.
 - b.) Enter root as a superuser:
At the ctuser prompt, type: **su -** ENTER
At the password prompt, type: **#bigguy** ENTER
 - c.) Insert the Class C software CD-ROM into the Console CD drive. Wait for the disk read to STOP.

- d.) Install the Class C software on the system:
At the superuser prompt, type: `install.classC` ENTER
When the installation process is complete, the Console reports the following:
Finished loading <System Type> Class C software
- e.) Verify that the Class C software is on the system:
At the superuser prompt, type: `showprods | grep -i classc`
Verify that the window displays a list of CLASSC file information and locations. For example (sample output):

```
I Helios_CLASSC_Product 09/18/2000 Helios OC Class C Product
26_2.8.2I_H2M3_H1.3M4
I Helios_CLASSC_Product.oc_image 09/18/2000 Files used on the Host
I Helios_CLASSC_Product.oc_image.g_subsys 09/18/2000 Files located on the
/usr/g partition
I Helios_H2_O2LESS_CLASSC_Product 09/18/2000 Helios H2.0 O2lessOC Class C
Product 226_2.8.2I_H2M3_H1.3M4
I Helios_H2_O2LESS_CLASSC_Product.oc_image 09/18/2000 Files used on the
Host
I Helios_H2_O2LESS_CLASSC_Product.oc_image.g_subsys 09/18/2000 Files
located on the /usr/g partition
```

- f.) Remove the Class C software CD-ROM from the Console CD drive.
- 2.) Install Service Key
Install the service key into the port on the front of the console.

3.17.2 Removing Class C Software

- 1.) On the Operator Console, open a Unix Shell window.
- 2.) Enter root as a superuser:
 - a.) At the ctuser prompt, type: `su -` ENTER
 - b.) At the password prompt, type: `#bigguy` ENTER
- 3.) Remove the Class C software from the system:
 - a.) At the superuser prompt, type: `uninstall.classC` ENTER
 - b.) When the un-install process is complete, the Console reports the following:
Class C software has been successfully removed from the system.
- 4.) Verify that the Class C software is no longer on the system:
 - a.) At the superuser prompt, type: `showprods | grep -i classc`
 - b.) Verify that the window DOES NOT display a list of CLASSC file information.

3.18 Class M (InSite) Software Installation

3.18.1 Load Class M Software

- 1.) On the Operator Console, open a Unix Shell window.
- 2.) Enter root as a superuser:
At the ctuser prompt, type: `su -` ENTER
At the password prompt, type: `#bigguy` ENTER
- 3.) Insert the Class M software CD-ROM into the Console CD drive. Wait for the disk read to STOP.
- 4.) Install the Class M software on the system:
At the superuser prompt, type: `install.classM` ENTER

When the installation process is complete, the Console reports the following:

Finished loading <System Type> Class M software

- 5.) Verify that the Class M software is on the system:
At the superuser prompt, type: **showprods | grep -i classm**
Verify that the window displays a list of CLASSM file information and locations.
- 6.) Remove the Class M software CD-ROM from the Console CD drive.

3.18.2 Removing Class M Software

- 1.) On the Operator Console, open a Unix Shell window.
- 2.) Enter root as a superuser:
 - a.) At the ctuser prompt, type: **su -** ENTER
 - b.) At the password prompt, type: **#bigguy** ENTER
- 3.) Remove the Class C software from the system:
 - a.) At the superuser prompt, type: **uninstall.classM** ENTER
 - b.) When the un-install process is complete, the Console reports the following:
Class M software has been successfully removed from the system.
- 4.) Verify that the Class M software is no longer on the system:
 - a.) At the superuser prompt, type: **showprods | grep -i classm**
 - b.) Verify that the window DOES NOT display a list of CLASSM file information.

Appendix A

System State Conversion

Section 1.0 Introduction

This procedure describes a system state conversion from IRIX to LINUX.

Section 2.0 Procedure

Do the following steps before powering down the Irix console and installing the GRE console.

- 1.) Save system state to an MOD on the Irix console before installing the new GRE/Linux console.
- 2.) Write down all of the system INFO information on the reconfig screens, including the network information. During the Load from Cold on the new console, you cannot restore the INFO file during the load.
- 3.) If the console has Connect Pro installed, write down the information when you run `installhisris` so it can be entered on the new console when installing the Connect Pro option.
- 4.) Power down the Irix console.
 - a.) Install the new GRE/Linux console. You will be using your existing SCIM/Keyboard, monitors, and mouse unless you received a new mouse. You will receive a new trackball that must be used on the new console. Your old trackball will no longer work. Install the new tower that has a DVD Ram drive installed and an MOD drive installed. When you power up the GRE/Linux console, make sure the power to the tower has been turned on (Switch on front of the tower) and the new trackball is installed. It plugs in to a cable near the connections for the keyboard and mouse. The extension cable for the trackball is already connected to the USB port at the rear of the HP computer.
 - b.) With the front cover off, note there are two places where you will be installing a CD during the LFC. The HP Host computer CD drive and the DARC CD drive. The OS and applications are loaded using the host computer CD. During the DARC load the OS is inserted in the DARC CD. The DARC application CD is loaded using the host computer CD.
- 5.) Start the LFC for the GRE/Linux console. When it asks if you want to restore the `INFO` file during the load, select NO.
- 6.) Complete the OS and Applications load and then reboot when it asks you to.
- 7.) Start up applications.
- 8.) When applications are up, select the Service Desktop and open a unix shell. Become `root` and load Class C & M. Close the shell.
- 9.) Open a new unix shell. You must be `ctuser`.
 - a.) Type `whoami` ENTER. The system should respond `ctuser`.
 - b.) Insert the Irix System State MOD into the tower MOD drive. The DVD drive should be empty.

-
- c.) Execute the following commands as *ctuser*.
`cd /usr/g/bin ENTER`
`sysstategui -MOD -IRX2LNX ENTER`
 - d.) When the system state GUI comes up, select ALL and then select RESTORE.
 - e.) The Irix to Linux conversion should take around 40 minutes. The command lines will scroll very quickly during the conversion in the window. When the scrolling stops, the system state has completed the conversion. When it has completed, close the window.
- 10.) In the same unix shell opened in step 9, execute the following commands as *ctuser*:
- a.) `mountMOD -I2L ENTER`
 - b.) `mv /usr/g/service/.bb /usr/g/service/bb.lfc ENTER`
 - c.) `mkdir /usr/g/service/.bb ENTER`
 - d.) `rm -rf /tmp/*bb ENTER`
 - e.) `cp /MOD/service_mod_data/system_state/*bb /usr/g/service/.bb/ENTER`
 - f.) `/usr/g/bin/SwapTubeData ENTER`
 - g.) `/bin/cp -f /tmp/*bb /usr/g/service/.bb/ ENTER`
- 11.) Remove the MOD from the drive.
- 12.) Put the option DVD into the DVD drive and install the options on the system.
- 13.) Reboot the system.
- 14.) Complete a flash download.

You should now be ready to scan.

Complete an Insite checkout. The system is a different model type, so you will have to have the insite center perform a checkout.

After insite has been checked out, save system state to DVD.

Appendix B

Example config/INFO Output

Section 1.0

Sample Output

This section contains an example system hardware configuration INFO file output that is displayed when you type the following in a shell: **cat /usr/g/config/INFO ENTER**

For some parameters, the values for your system may differ from those in the example.

```
setenv SERVICE_ID GE_SERVICE
setenv HOSPITAL_NAME _____
setenv MENUTZ "America" _____
setenv SUNTZ "US/Central" _____
setenv TZ "CST6CDT US" _____
setenv TZTEXT "Chicago" _____
setenv REGENERATE_DATABASE no
setenv NEXT_MOD_NUMBER 1
setenv NEXT_DIAGNOSTIC_EXAM_NUMBER 50000
setenv GATEWAY_HOSTNAME _____
setenv GATEWAY_IP _____
setenv GATEWAY_NETMASK _____
setenv GATEWAY_BROADCAST _____
setenv GATEWAY_DEFAULT _____
setenv TUBETYPE 15 (Performix) / 16 (Hercules) / 17 (OEM) _____
setenv PDUTYPE CT_Compact / NG PDU _____
setenv DASMTYPE 2
setenv KV_VALUES "80 100 120 140" _____
setenv DEFAULTARCHIVE DICOM
setenv ADVANTAGERW FALSE
setenv CTIDATETIMEFORMAT 4
setenv TIMESEPAR /
setenv WEIGHT lb
setenv DOCTITLE "radiologist"
setenv CAMERATYPE ""
setenv SUITEID _____
setenv FILTERTYPE 2
setenv CAMERA no
setenv DEFFORMAT 12
setenv DOGREYSCALE yes
setenv DOSLIDES no
```

```
setenv DOZOOM no
setenv FILMINTERP sharp
setenv PRINTER_HOSTNAME _____
setenv PRINTER_IPADDRESS _____
setenv PRINTER no
setenv O2_PRESENT n
setenv GANTRY_TYPE h2 / hp/ wb _____
setenv LANGUAGE en_US
setenv DOSE y
setenv NETWORK yes
setenv USENIS yes
setenv CHGDARCSUBNET yes
setenv DOMAINNAME _____
setenv NIS_SERVER _____
setenv DARC_SUBNET _____
setenv NEXT_PATIENT_EXAM_NUMBER 1
setenv MOBILESYSTEM no _____
setenv STREAK y
setenv INROOMSCAN off
setenv AAA n
setenv AUTOMA_INDEX_TABLE_TYPE type2 _____
setenv RECONHARDWARE GRE / No O2 Pegasus _____
setenv DASTYPE SDAS or MDAS 4 / 8/ 16 _____
setenv DETECTORTYPE Warp3
setenv REFRESH_RATE 76

setenv yp s
setenv ypmaster ${SUITEID}_OC0
setenv HOSTNAME SBC0
setenv ALIAS ${SUITEID}_SBC0
setenv UNSYSID $GATEWAY_HOSTNAME
setenv SERVID $GATEWAY_HOSTNAME
setenv tubeType $TUBETYPE
setenv ACRNEMA xxxx
setenv REGEN no

setenv HRCTTITLE _____
setenv HRPORNUM _____
setenv HRAETITLE _____
setenv HRIPADDR _____
{ctuser@hosatname}
```

Appendix C

DARC OS Load Pop-Up Boxes

Section 1.0

Introduction

This section describes pop-up message boxes that may appear during a DARC OS load for Version 1.7.10, and provides additional information on what to do if you get one.

This section applies ONLY to products using DARC/Host OS version 1.7.10.

LightSpeed 1.X (LightSpeed QX/i)

LightSpeed 2.X (LightSpeed Plus/QX/i)

LightSpeed 3.X (LightSpeed Ultra/Plus) and HiSpeed QX/i M4 and later

LightSpeed 4.X (LightSpeed 16) M4

LightSpeed 5.X (Pro16 (100kW)) M3.1

LightSpeed 5.X (Pro16 (80kW)) M3

LightSpeed 5.X (RT) M3

LightSpeed 5.x Plus

Section 2.0 Procedure

Locate the appropriate pop-up box in the following subsections and refer to the action to take.

2.1 Error - Cannot connect with DARC

POP-UP MESSAGE

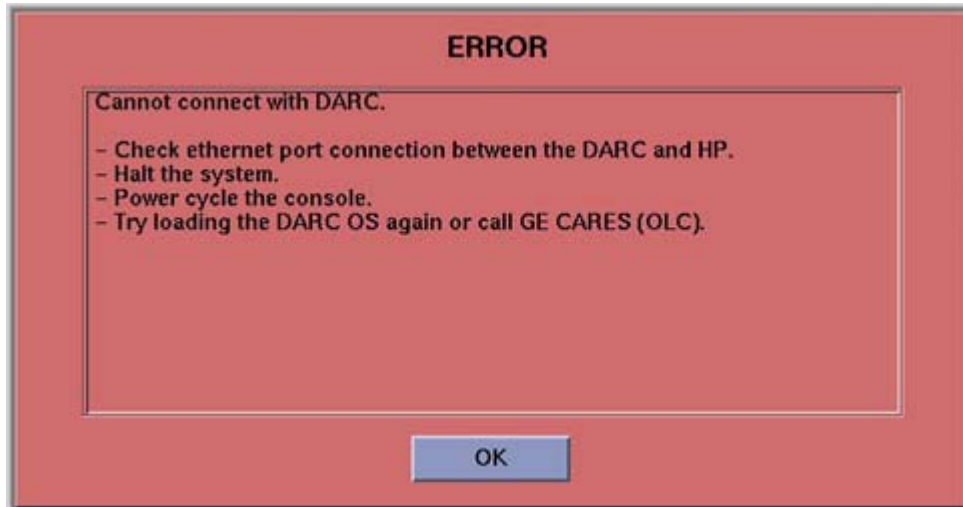


Figure C-1 Cannot Connect With DARC

POP-UP MESSAGE DESCRIPTION

SOL (Serial Over LAN) checks out good but a failure occurs.

ACTION TO TAKE

- 1.) At the rear of the DARC, verify Ethernet 3 to HP Host PC Ethernet C is installed properly.
- 2.) Click OK.
- 3.) Open a Unix Shell.
- 4.) Type: `halt` ENTER.
- 5.) When console message states power down then Power Off the Operator Console.
- 6.) Wait 30 seconds and Power On the Operator Console.
- 7.) Try the DARC OS load again.

2.2 Error - BMC connection on DARC node is not responding

POP-UP MESSAGE



Figure C-2 BMC Connection on DARC Node is Not Responding

POP-UP MESSAGE DESCRIPTION

Cannot connect to the DARC or an internal error in the DARC.

ACTION TO TAKE

- 1.) At the rear of the DARC, verify Ethernet 3 to HP Host PC Ethernet C is installed properly.
- 2.) Click OK.
- 3.) Open a Unix Shell.
- 4.) Type: `halt` ENTER.
- 5.) When console message states power down, then Power Off the Operator Console.
- 6.) Wait 30 seconds and Power On the Operator Console.
- 7.) Try the DARC Node OS load again.
- 8.) Try new DARC Node to HP Ethernet cable.
- 9.) Call OLC before replacing DARC.

2.3 Error - Cannot send boot command to DARC

POP-UP MESSAGE



Figure C-3 Cannot Send Boot Command to DARC

POP-UP MESSAGE DESCRIPTION

During initial DARC Operating System load process something goes wrong.

ACTION TO TAKE

- 1.) Remove OS CD from DARC drive.
- 2.) Click OK.
- 3.) Open a Unix Shell.
- 4.) Type: `halt` ENTER.
- 5.) When console message states power down then Power Off the Operator Console.
- 6.) Wait 30 seconds and Power On the Operator Console.
- 7.) Try the DARC Node OS load again.

2.4 SOL Service - Run the Serial Over LAN Service CD

POP-UP MESSAGE

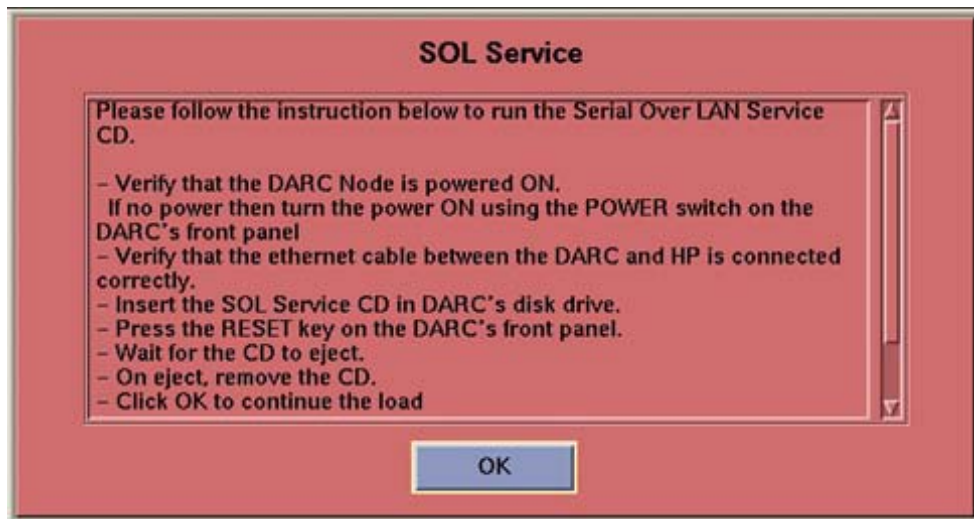


Figure C-4 Run the Serial Over LAN Service

POP-UP MESSAGE DESCRIPTION

DARC Internal BMC chip is faulty or vendor failed to flash the IP Address.

ACTION TO TAKE

Follow instructions in pop-up box using the LightSpeed Xstream Console Serial Over Lan Service CD (P/N 2399297).

2.5 Error - Cannot send a complete signal to DARC

POP-UP MESSAGE

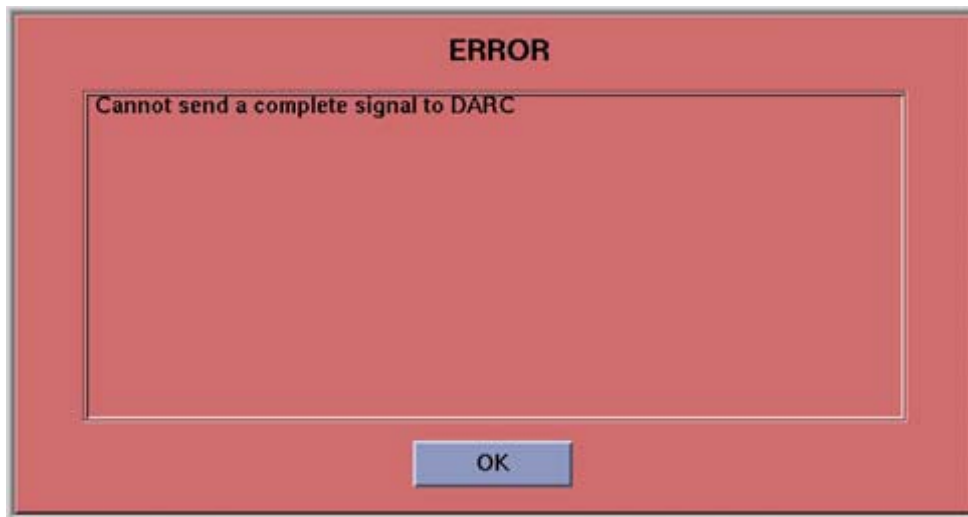


Figure C-5 Cannot send complete signal to DARC

POP-UP MESSAGE DESCRIPTION

This is a Pop-Up that is used in conjunction with the informational pop-ups listed in [Section 2.6](#). It notifies the user an ERROR has occurred.

ACTION TO TAKE

1. Click on OK.
2. Refer to [Section 2.6](#).

2.6 Information - DARC Node loaded OS but complete signal not sent

POP-UP MESSAGE



Figure C-6 DARC Node OS Loaded But No Complete Signal Sent (Tips Ref)



Figure C-7 DARC Node OS Loaded But No Complete Signal Sent

POP-UP MESSAGE DESCRIPTION

Both messages are essentially the same. Ignore the "tips and work around manual" reference. The DARC Operating System was properly loaded properly, however, the DARC "complete" signal was lost or missed within the timeout period.

ACTION TO TAKE

- 1.) At the DARC front panel, press the RESET Switch . See [Figure C-8](#).



Figure C-8 DARC Reset Switch Location

- 2.) Eject the CD.
- 3.) The Pop-Up box in the LFC procedure may pop-up. Click on OK.
- 4.) Verify the OS was successfully loaded per the LFC procedure:
 - a.) Open a Unix Shell.
 - b.) {ctuser@hostname} **su** SPACE - ENTER
 - c.) Password: #bigguy ENTER
 - d.) [root@hostname]# **rsh** SPACE darc ENTER
Last login: Thu Oct 9 11:17:35 on ttyS1
You have new mail.
 - e.) [root@localhost root] **exit** ENTER <- The prompt shows the OS was loaded properly.
 - f.) Close the Unix Shell.
- 5.) Click on OK.
- 6.) Continue with software install. Load DARC applications on the HP Host Computer next.

2.7 DARC Node - Cable Checkout

- 1.) Verify the rear power cable is connected, switch is turned on and fan is operating.
- 2.) Verify front power LED is illuminated.
- 3.) Verify Ethernet cable is in good condition and connected as follows:
 - Ethernet port C --- DARC Node Ethernet port 3
 - HP (right-card - top Ethernet port)



Figure C-9 DARC Node eth3- Bottom Connection Between SCSI and Keyboard

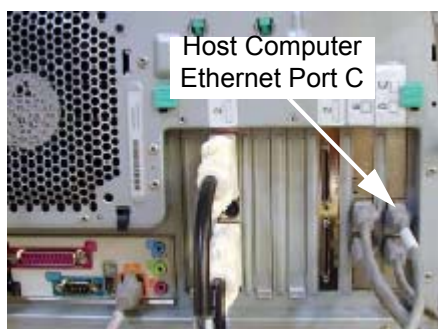


Figure C-10 HP Host Computer Ethernet Port C (See Cable Label)

2.8 DARC Node - Remote Shell

This subsection will verify the DARC Node remote shell can be invoked to determine if the DARC Node is up. The prompt `root@localhost` lets the user know the DARC OS has been loaded successfully.

Open a Unix Shell and type the following:

```
{ctuser@hostname} su SPACE - ENTER
Password: #bigguy ENTER
[root@hostname]# rsh darc ENTER
Last login: Thu Oct  9 11:17:35 on ####
You have new mail.
[root@localhost root ]# exit ENTER
logout
rlogin: connection closed.
[root@hostname]#
```

2.9 DARC Node --- Serial Over LAN Checkout

Verification of the Serial Over LAN check is almost always the initial test performed on the DARC after power and cable verification. It has been placed after ping and remote shell checkout because they are very quick tests. With SOL Checkout the telnet connection is made and the SOL connection from the HP Host Computer to the DARC Node is then established. This test will verify the DARC Node was booted and is up.

Open a Unix Shell and type the following:

```
{ctuser@hostname} su SPACE - ENTER
Password: #bigguy ENTER
[root@hostname]# telnet localhost 623 ENTER
Trying 127.0.0.1...
Connected to localhost. <- This output indicates HP Host Apps were loaded and the
cliservice proxy server is running.
Escape character is '^]'.
Server: darc ENTER
Username: ENTER
Password: ENTER
Login successful <- This output indicates the SOL connection is established.
dpccli> console ENTER

GEMS Linux 1.7.10 <- This is the Console Operating System Software Revision.
See you on the darc side of the moon.* <- This indicates the DARC Node is up.
darc login: root ENTER *** See note below. ***
Password: #bigguy ENTER
Last login: Thu Oct  9 11:11:15 on ttyS1
You have new mail.
[root@darc ~]# ~.(<- Enter a "tilde" "period" to terminate the telnet.

dpccli> exit ENTER
Connection closed by foreign host.
[root@hostname]#
```

Note: If the darc login prompt is not present and localhost, localdomain prompt is displayed, then try reloading the DARC Node OS and DARC Application software per the procedure. The problem could be that the DARC Apps did not load correctly. Always be sure the DARC OS matches the HP Host OS already loaded (example, if the applications software requires Console Operating System CD version 1.7.10, then both the DARC and HP Host must be loaded with this OS version).

Appendix D

General Information

Section 1.0 Introduction

This Appendix contains the following "how to" procedures that may be useful during a software load:

- Remote Shell — Check Internal Network
- HP Computer CDROM — Mount and Unmount Process
- Peripheral Tower DVD — Mount and Unmount Process
- Peripheral Tower DVD — Initializing a DVD
- Peripheral Tower MOD — Mount and Unmount Process

Section 2.0 Procedures

2.1 Remote Shell — Check Internal Network

- To become root within a Unix Shell:
 - 1.) Type: `su -` ENTER
 - 2.) Type the root password: `#bigguy` ENTER
- To check the internal network as root, type: `rsh darc` ENTER
 - 1.) Type: `exit` to close root@darc shell, or to check IG1, type: `rsh ig1` ENTER
 - 2.) Type: `exit` to close root@IG1 shell and `exit` to close root@darc shell
 - 3.) Close the Unix Shell.

2.2 HP Computer CDROM — Mount and Unmount Process

- 1.) Insert the CD into the HP Host Computer DVD ROM drive
- 2.) Open up a Unix Shell and type these commands:
 - a.) ctuser: `su -` ENTER
 - b.) password: `#bigguy` ENTER
 - c.) root: `mount /mnt/cdrom` ENTER
- 3.) The CD has now been mounted. You can see what is on it by typing:
`ls /mnt/cdrom` ENTER
- 4.) You must unmount the CD in order to remove it from the drive; do this by typing:
`umount /mnt/cdrom` ENTER

2.3 Peripheral Tower DVD — Mount and Unmount Process

- 1.) Insert the DVD into the Peripheral Tower DVD RAM drive
- 2.) Open up a Unix Shell and type: `mountDVD` ENTER
- 3.) Now the DVD has been mounted. You can see what is on it by typing:
`ls /DVD` ENTER
- 4.) You must unmount the DVD in order to remove it from the drive; do this by typing:
`unmountDVD` ENTER

2.4 Peripheral Tower DVD — Initializing a DVD

If a DVD needs to be initialized on new media, open a Unix Shell and perform the following:

- 1.) Type: `su -` ENTER
- 2.) Enter the root password: `#bigguy` ENTER
- 3.) Type: `mountDVD` ENTER
- 4.) Type: `mkfsDVD` ENTER; answer `y` for yes and close the unix shell.

2.5 Peripheral Tower MOD — Mount and Unmount Process

- 1.) Insert the MOD into the Peripheral Tower MOD drive
- 2.) Open up a Unix Shell and type: `mountMOD` ENTER
- 3.) Now the MOD has been mounted. You can see what is on it by typing:
`ls /MOD` ENTER
- 4.) You must unmount the MOD in order to remove it from the drive; do this by typing:
`umountMOD` ENTER

Note: If MOD cannot be unmounted, select: IMAGE WORKS > ARCHIVE > DETACH.



GE MEDICAL SYSTEMS

***GE MEDICAL SYSTEMS-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.***

***GE MEDICAL SYSTEMS-EUROPE: FAX 33.1.40.93.33.33
PARIS, FRANCE***

GE MEDICAL SYSTEMS-ASIA: FAX 65.291.7006